

END TERM EXAMINATION

FIRST SEMESTER [MCA] DECEMBER 2024

Subject: Computer Networks

Paper Code: MCA-103

(BATCH 2020 ONWARDS)

Maximum Marks: 60

Time: 3 Hours

Note: Attempt any five questions in all including Q.No.1 which is compulsory. Attempt one question from each unit.

Q1 Answer the following questions, briefly:

(2×10=20)

- Compare and Contrast TCP/IP model with OSI model of computer network.
- With a suitable example, enlist the responsibilities of Data Link layer.
- Compare IPv4 and IPv6 headers.
- List the difference in the sliding window used for flow control at data link layer and transport layer.
- Describe the components of mobile IP.
- Define autonomous system
- Illustrate the topology used for Bluetooth wireless technology
- Describe dynamic web pages.
- Enlist the different type of messages in ICMPv6
- In TCP, if the value of HLEN is 0111, how many bytes of options are included in the segment?

UNIT-I

- Q2 a) Discuss the advantages of optical fiber over twisted pair and coaxial cable. Also explain its mode and the purpose of cladding in optical fiber. (5)
- b) Sketch the Manchester and Differential Manchester encoding for the bit stream: 1010110000110100. (5)

OR

- Q3 a) Explain the Analog-to-analog conversion technique by considering an appropriate example. (5)
- b) Five channels, each with a 100-KHz bandwidth, are to be multiplexed together. Determine the minimum bandwidth of the link if there is a need for a guard band of 10KHz between the channels to prevent interference? Also, differentiate between time division multiplexing and frequency division multiplexing. (5)

UNIT-II

- Q4 a) Explain the purpose and detail working of CSMA/CD protocol. (5)
- b) Determine the data sent from sender side using CRC generator polynomial x^4+x+1 when the original data frame is: 1101011011 (5)

P.T.O.

OR

- Q5 a) Illustrate the Vulnerable time for pure aloha random access protocols. A pure aloha network transmits 200-bit frames on a shared channel of 200Kbps. Determine the requirement to make this frame collision free. (5)
- b) Differentiate between circuit switching, message switching and packet switching. (5)

UNIT-III

- Q6 a) Differentiate between static routing and dynamic routing algorithm. (5)
- b) List the different classes of addresses used in IPv4, specify their host ID and network ID's. List their ranges in dotted decimal notation. (5)

OR

- Q7 a) A company is granted a site address 201.70.64.0. The company needs six subnets. Design the subnets (Subnet masks for each subnet) starting and ending address of each subnet. (5)
- b) Discuss in detail an interior gateway routing protocol that uses Bellman-Ford Algorithm to find the optimal route. (5)

UNIT-IV

- Q8 a) Illustrate TCP header. Emphasis upon the flags which will be used in establishing a connection using a three-way handshake. (5)
- b) Discuss different scenarios in which email can be sent from sender to receiver. (5)

OR

- Q9 Write short notes on the following (5+)
- DNS
 - Connectionless protocol in transport layer

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END TERM EXAMINATION

First Semester (MCA) January 2024

Paper Code: MCA-103

Subject: Computer Networks

Time: 3 Hours

Maximum Marks: 60

Note: Attempt five questions in all including Q.No.1 which is compulsory.
Select one question from each unit.

Q1 Attempt questions (any four): [3x4=12]

- What do you understand with digital signal encoding (digital to digital encoding) techniques? Why these techniques are needed? convert the following bit stream into a digital signal using Manchester encoding technique.
101000010
- Compare wired and wireless transmission mediums in terms of speed, security and scale. Explain the hidden station and exposed station problems with proper diagrams.
- Differentiate between CSMA and CSMA/CD.
- What is CIDR? How does it solve the problem of shortage of IPv4 addresses?
- Explain the technique of Checksum calculation with proper example.

UNIT-I

- Q2 a) What is analog to digital encoding? What is its use in data communication? Explain the PCM technique in detail. (7)
b) What is MODEM? How does it make the communication possible between two remote machines? (5)
- Q3 Explain the layered architecture? Why Network models have been designed using layered architecture? Explain layers of TCP/IP model in detail. (12)

UNIT-II

- Q4 a) Differentiate between error detection and retransmission technique and error detection and correction techniques. Explain the scenarios in which these techniques are preferred. Encode following data word using Hamming code with even parity. (7)
1001001000011101
- b) Explain Stop and Wait ARQ with an example and required diagrams. (5)
- a) Explain Ethernet frame format in detail. What is the need of padding in it? (7)
- b) Explain DCF Medium Access Control (MAC) method of wireless LAN networks in detail. (5)

UNIT-III

- a) Explain the IPv4 address classes with their features. How address distribution was done in classful addressing method? Explain. (5)
- b) What is link state routing algorithm? Explain the method of routing table population in link state routing. (7)

P.T.O.

Q7

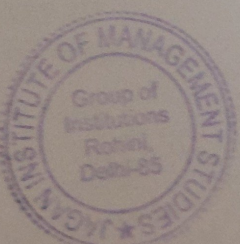
- Explain IPv6 address auto configuration process. What happens when the autoconfiguration process gets fail.
- Explain the Multicast routing protocols based on source-based and group shared tree methods.

UNIT-IV

- Q8 a) Explain TCP protocol header in detail.
b) Mention the applications where UDP protocol is used with proper reason.

Q9

- Explain the process of mail transfer including all the necessary components of the process. Include proper diagrams if needed.
- Explain Private and public key cryptography with their Pros and Cons. For large amount of data transfer which one would be preferred?



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Roll No.)
END TERM EXAMINATION

Exam Roll No. 2073010422

Paper Code: MCA 103

FIRST SEMESTER [MCA] JANUARY-FEBRUARY 2023

Time: 3 Hours

Subject: Computer Networks

Maximum Marks: 75

Note: Attempt all questions as directed. Internal choice is indicated.

- Q1. Answer the following briefly: (2.5x10=25)
- (a) What are the different types of transmission media?
 - (b) Explain the difference between node-to-node delivery and source-to-destination delivery?
 - (c) What are the advantages of Piggy Backing?
 - (d) Difference between Bandwidth and Throughput.
 - (e) Describe the count-to-infinity problem.
 - (f) What is the purpose of cladding in optical fiber?
 - (g) Differentiate RST and FIN flag in TCP header.
 - (h) Differentiate Connectionless and Connection Oriented services.
 - (i) Differentiate between Static Routing and Dynamic Routing algorithm with examples?
 - (j) Distinguish between bit rate and baud rate.

UNIT-I

- Q2. (a) Explain Pulse Code Modulation Techniques. (6.5)
(b) What are the different techniques to convert Analog-to-digital? Explain the process. (6)

OR

- Q3. (a) Give the NRZ-I, NRZ-I, Manchester, Differential Manchester, and Bipolar encoding of the digital data: 00110101000111. (6.5)
(b) Explain Soft handoff and Hard handoff. (6)

UNIT-II

- Q4. (a) What are the different types for Error Detection Techniques? Explain. (6.5)
(b) Difference between CSMA, CSMA/CD and CSMA/CA. (6)

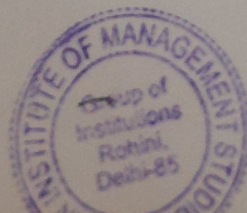
OR

- Q5. (a) Explain Stop-and-Wait ARQ protocol to the Go-Back-N ARQ Protocol. (6.5)
(b) Given a 10 bits sequence 11011011100 and a divisor is 1101, find the CRC. Also check your result at receiver side. (6)

UNIT-III

- Q6. (a) Explain the architecture of Bluetooth. Compare PICONET and SCATTERNET. (6.5)

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- (b) What are the different UNICAST routing algorithms and explain the difference between distance vector and link state routing algorithms. (6)

OR

- Q7. (a) Explain IP MULTICAST routing protocols? (6.5)
(b) Explain IPv6 Extension Header. What are the various techniques for transition from IPv4 to IPv6? (6)

UNIT-IV

- Q8. (a) Explain TCP header. (6.5)
(b) Explain Domain Name Server (DNS)? Explain the address resolution techniques in DNS. (6)

OR

- Q9. (a) Explain the Silly Window Syndrome. Also, explain the cause of Silly Window Syndrome from Sender and Receiver Side. (6.5)
(b) Explain POP and IMAP in email system. (6)

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